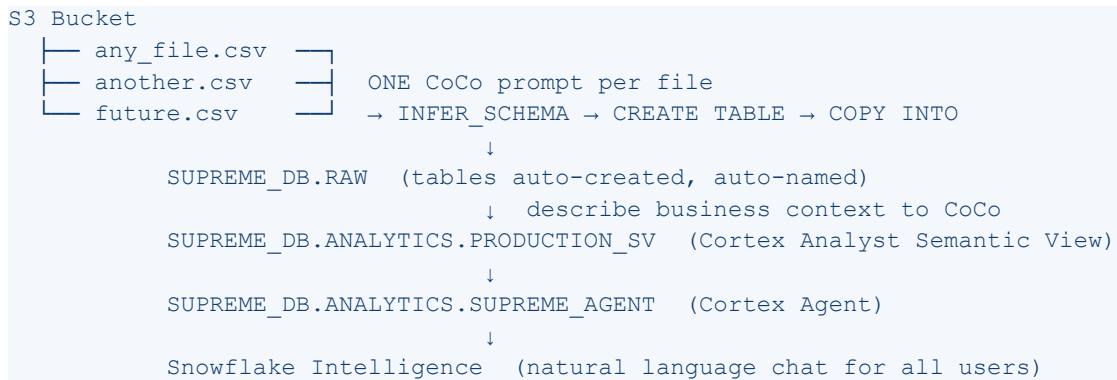


Data to Intelligence — Cortex Code (CoCo) Guide

Any CSV from S3 → Snowflake → Cortex Agent → Snowflake Intelligence

End-to-End Architecture



Section 1 — One-Time Setup

This section is done **once**. After completing it, you never touch it again — not when new files arrive, not when schemas change, not when new data sources are added.

Step 1.1 — Create Database, Schemas, and File Format

One prompt creates everything needed: the database, two schemas, and a named CSV file format that will be reused for all future file loads.

CoCo Prompt

Create a database called SUPREME_DB with two schemas: RAW for ingested data and ANALYTICS for the semantic view and agent. Also create a named CSV file format called CSV_AUTO in SUPREME_DB.RAW — it should handle quoted fields, auto-detect dates, treat empty strings as NULL, and skip the header row.

CoCo runs:

```
CREATE DATABASE IF NOT EXISTS SUPREME_DB;
CREATE SCHEMA IF NOT EXISTS SUPREME_DB.RAW;
CREATE SCHEMA IF NOT EXISTS SUPREME_DB.ANALYTICS;

-- Named format: reused for every file load, forever
CREATE OR REPLACE FILE FORMAT SUPREME_DB.RAW.CSV_AUTO
  TYPE = CSV
  FIELD_OPTIONALLY_ENCLOSED_BY = '"'
  SKIP_HEADER = 1
  NULL_IF = ('NULL', 'null', 'N/A', '')
  EMPTY_FIELD_AS_NULL = TRUE
  DATE_FORMAT = 'AUTO'
  TIMESTAMP_FORMAT = 'AUTO'
  TRIM_SPACE = TRUE;
```

Why a named file format? It is referenced in every `INFER_SCHEMA` and `COPY INTO` call. If you ever need to change parsing behaviour (e.g. different date format), you update it in one place and every future load picks up the change automatically.

Step 1.2 — Create Storage Integration (S3 → Snowflake)

The storage integration is a secure, credential-free link between Snowflake and your S3 bucket. It is created once and works for every file you ever upload to that bucket.

CoCo Prompt

Create a storage integration called SUPREME_S3_INT that allows Snowflake to read from S3 bucket s3://supreme-data-bucket/. My AWS IAM role ARN is arn:aws:iam::ACCOUNT_ID:role/SnowflakeRole.

```
CREATE STORAGE INTEGRATION SUPREME_S3_INT
  TYPE = EXTERNAL_STAGE
  STORAGE_PROVIDER = 'S3'
```

```
ENABLED = TRUE
STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::ACCOUNT_ID:role/SnowflakeRole'
STORAGE_ALLOWED_LOCATIONS = ('s3://supreme-data-bucket/');
```

Action required — AWS trust policy:

After creating the integration, CoCo will display `STORAGE_AWS_IAM_USER_ARN` and `STORAGE_AWS_EXTERNAL_ID`. Paste these into the trust policy of your AWS IAM role. Use this prompt:

CoCo Prompt

Show me the AWS IAM trust policy JSON I need for SUPREME_S3_INT. I will paste it directly into the AWS IAM console.

Step 1.3 — Create External Stage

The stage is the named pointer to your S3 path. Any file you drop into this S3 location becomes immediately accessible to Snowflake.

CoCo Prompt

Create an external stage called SUPREME_DB.RAW.S3_STAGE pointing to s3://supreme-data-bucket/data/ using SUPREME_S3_INT and file format SUPREME_DB.RAW.CSV_AUTO.

```
CREATE STAGE SUPREME_DB.RAW.S3_STAGE
  STORAGE_INTEGRATION = SUPREME_S3_INT
  URL                 = 's3://supreme-data-bucket/data/'
  FILE_FORMAT         = SUPREME_DB.RAW.CSV_AUTO;
```

One-time setup is now complete. You have: a database, two schemas, a reusable CSV format, a storage integration, and a stage. Everything from here on uses these four objects — you will never need to recreate them.

Section 2 — Load Any CSV File

This is the repeatable pattern. It works for **any CSV file** regardless of schema. The table is created automatically from the file's own headers. You never need to know the column names or types in advance.

How it works under the hood: ① `INFER_SCHEMA` reads the CSV headers and detects data types. ② `CREATE TABLE USING TEMPLATE` creates a table whose columns exactly match the file. ③ `COPY INTO ... MATCH_BY_COLUMN_NAME` loads the data matching by header name — column order does not matter, extra columns in future file versions are handled automatically.

The Universal Load Prompt

This single prompt is used for every file — first load, future updates, entirely new file types.

CoCo Prompt

Load the file [filename.csv] from @SUPREME_DB.RAW.S3_STAGE into SUPREME_DB.RAW. Automatically detect the schema, create the table if it does not exist, and load the data. Use the filename (without extension) as the table name. Show me the column names, types, and row count after loading.

CoCo executes this sequence automatically:

```
-- ① Detect schema from the file itself
SELECT * FROM TABLE(
  INFER_SCHEMA(
    LOCATION    => '@SUPREME_DB.RAW.S3_STAGE/filename.csv',
    FILE_FORMAT => 'SUPREME_DB.RAW.CSV_AUTO'
  )
);

-- ② Create table whose columns exactly match the CSV headers
CREATE TABLE IF NOT EXISTS SUPREME_DB.RAW.FILENAME
USING TEMPLATE (
  SELECT ARRAY_AGG(OBJECT_CONSTRUCT(*))
  FROM TABLE(
    INFER_SCHEMA(
      LOCATION    => '@SUPREME_DB.RAW.S3_STAGE/filename.csv',
      FILE_FORMAT => 'SUPREME_DB.RAW.CSV_AUTO'
    )
  )
);

-- ③ Load data — matched by column name, not position
COPY INTO SUPREME_DB.RAW.FILENAME
FROM      @SUPREME_DB.RAW.S3_STAGE/filename.csv
FILE_FORMAT = SUPREME_DB.RAW.CSV_AUTO
MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE
ON_ERROR = 'CONTINUE';

-- ④ Confirm
SELECT COUNT(*) AS row_count FROM SUPREME_DB.RAW.FILENAME;
```

`MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE` is the key setting that makes this file-independent. It ignores column order and matches solely by header name — so if a future version of the same file has columns in a different order, or adds new columns, the load still works correctly.

Loading Your Current Files (First Time)

For Supreme's initial set of files, a single batch prompt loads all of them:

CoCo Prompt

Load all of the following CSV files from @SUPREME_DB.RAW.S3_STAGE into SUPREME_DB.RAW. For each file: auto-detect schema, create the table, load the data, show row count.

Files:

- 23-24 sales.csv
- 24-25 sales.csv
- 25-26 sales.csv
- plant_capacity.csv
- machine_available.csv
- pending_sto.csv

Use the filename (without extension, spaces replaced with underscores) as the table name.

After loading, CoCo shows a summary like:

Table	Columns	Rows Loaded	Status
RAW.SALES_23_24	4	163,027	OK
RAW.SALES_24_25	4	172,168	OK
RAW.SALES_25_26	4	178,080	OK
RAW.PLANT_CAPACITY	4	110	OK
RAW.MACHINE_AVAILABLE	2	8	OK
RAW.PENDING_STO	2	58	OK

Loading a New File in the Future

When a new CSV arrives — whether it is a new fiscal year of sales, an updated STO snapshot, or a completely new type of data — the prompt is identical:

CoCo Prompt

Load the file production_actuals_apr2026.csv from @SUPREME_DB.RAW.S3_STAGE into SUPREME_DB.RAW. Auto-detect schema, create the table, load data, show row count.

CoCo handles it end-to-end. No schema knowledge required. After loading, skip to Step 3.3 to add the new table to your semantic view.

When a File's Schema Changes (New Columns)

If a future version of a file has additional columns, tell CoCo:

CoCo Prompt

A new version of plant_capacity.csv has arrived. Load it into the existing PLANT_CAPACITY table — add the new columns first, then load the data. Show me what changed.

CoCo will:

- Run `INFER_SCHEMA` on the new file to detect columns
 - Compare against the existing table structure
 - Run `ALTER TABLE ADD COLUMN` for any new columns
 - Load the new file with `MATCH_BY_COLUMN_NAME` — existing rows are unaffected
-

Diagnosing Load Errors

CoCo Prompt

Some rows failed to load from [filename.csv]. Show me the error details and how many rows were affected.

CoCo queries `COPY_HISTORY` and surfaces each error with the line number and reason. Common issues: date format not matching AUTO, a numeric column containing text, encoding problems. CoCo will suggest the specific fix.

Section 3 — Semantic Layer

The semantic view is the intelligence layer. It translates raw column names into business language, defines how tables join, calculates metrics, and gives the LLM the context it needs to answer questions accurately. A well-described semantic view is the difference between an agent that gives vague answers and one that answers like a domain expert.

Step 3.1 — Describe Your Tables to CoCo

Before building anything, describe your data in plain English. CoCo will reflect the understanding back — correct any errors at this stage, not after the semantic view is built.

CoCo Prompt

I am going to describe the tables in SUPREME_DB.RAW. Confirm your understanding before we build a semantic view.

SALES_23_24, SALES_24_25, SALES_25_26: These three tables are sales transactions for fiscal years 2023-24, 2024-25, and 2025-26. Each row is one invoice line. QTY_BASE_UNIT is the quantity in pieces (NOS). Negative values are sales returns. Net sales = SUM of all quantities including negatives. MATERIAL_CODE is the product identifier and is the join key to other tables.

PLANT_CAPACITY: Each row is a machine-mould-material combination on the production floor. MACHINE_TYPE is the press type (e.g. FM250, FM300). MOULD_NUMBER is the tool fitted to the machine. CAPACITY_PER_DAY is the maximum units this combination can produce in one day. MATERIAL_CODE joins to the sales tables.

MACHINE_AVAILABLE: AVAILABLE_MACHINES is the physical count of each MACHINE_TYPE on the production floor. Joins to PLANT_CAPACITY on MACHINE_TYPE.

PENDING_STO: Each row is an unfulfilled stock transfer order. PENDING_QTY is the quantity not yet dispatched. MATERIAL_CODE joins to the sales tables.

Summarise your understanding so I can confirm before we proceed.

Correct CoCo if anything is wrong. Examples:

"No — CAPACITY_PER_DAY is for the mould, not the machine type overall."

"The STO table has a snapshot date column — it represents the state as of that date, not live."

Each correction you make now is baked into the semantic view permanently.

Step 3.2 — Create the Semantic View

CoCo Prompt

Create a Cortex Analyst semantic view called SUPREME_DB.ANALYTICS.PRODUCTION_SV across all tables in SUPREME_DB.RAW.

Treat SALES_23_24, SALES_24_25, SALES_25_26 as three partitions of the same fact — query them as a unified UNION-like source with a derived FISCAL_YEAR dimension.

Join keys:

- Sales tables → PLANT_CAPACITY via MATERIAL_CODE
- Sales tables → PENDING_STO via MATERIAL_CODE
- PLANT_CAPACITY → MACHINE_AVAILABLE via MACHINE_TYPE

Business term mappings:

- "material" or "product" = MATERIAL_CODE
- "machine" = MACHINE_TYPE
- "mould" = MOULD_NUMBER
- "net sales" = SUM(QTY_BASE_UNIT) including negatives
- "gross sales" = SUM where QTY > 0
- "returns" = ABS(SUM where QTY < 0)
- "daily capacity" = CAPACITY_PER_DAY
- "total plant capacity" = SUM(CAPACITY_PER_DAY * AVAILABLE_MACHINES)
- "backlog" or "pending STO" = SUM(PENDING_QTY)

Metrics to define:

1. net_sales_qty — SUM of QTY_BASE_UNIT across all fiscal years
2. gross_sales_qty — positive quantities only
3. return_qty — ABS of negative quantities
4. return_rate_pct — return_qty / gross_sales_qty * 100
5. total_capacity_per_day — SUM(CAPACITY_PER_DAY * AVAILABLE_MACHINES)
6. pending_sto_qty — SUM(PENDING_QTY)
7. capacity_utilisation_pct — net_sales_qty / total_capacity_per_day * 100

Verified queries to seed:

- "What are the top 10 materials by net sales in FY 25-26?"
- "Which machine type has the highest total daily capacity?"
- "Which materials have a return rate above 5%?"
- "What is the pending STO backlog by material?"

Step 3.3 — Validate the Semantic View

CoCo Prompt

Test PRODUCTION_SV with these 6 questions. Show the generated SQL and result for each:

1. What is the net sales quantity for each material in FY 25-26?
2. Which machine type has the highest total daily capacity?
3. Which materials have the highest return rate?
4. What is the total pending STO backlog?

5. How does net sales in FY 25-26 compare to FY 24-25?
6. Which materials have pending STO greater than their daily capacity?

For any question that gives a wrong answer, use this prompt:

CoCo Prompt

PRODUCTION_SV returned [wrong answer] for the question "[question]". The correct answer is [correct answer / explanation]. Update the semantic view to fix this.

Step 3.4 — Adding a New Table to the Semantic View

When a new file is loaded and its table exists in `SUPREME_DB.RAW`, extend the semantic view with one prompt.

CoCo Prompt

I just loaded a new table `SUPREME_DB.RAW.PRODUCTION_ACTUALS` which has columns: `PRODUCTION_DATE`, `MACHINE_TYPE`, `MATERIAL_CODE`, `ACTUAL_QTY_PRODUCED`. Each row is the actual quantity produced by a machine for a material on a given day. Add this table to `PRODUCTION_SV`, join it to `PLANT_CAPACITY` on `MACHINE_TYPE` and `MATERIAL_CODE`, and define a new metric: `actual_vs_capacity_pct = actual_qty_produced / total_capacity_per_day * 100`.

Section 4 — Cortex Agent, Chat Window, Model Switching

Step 4.1 — Create the Agent

CoCo Prompt

Create a Cortex Agent called `SUPREME_DB.ANALYTICS.SUPREME_AGENT`.

Instructions:

- Answer questions about sales performance, machine capacity, mould configurations, and pending stock transfer orders for Supreme Industries
- Users are production managers and sales analysts — keep answers in business language
- When answering about "sales", clarify whether the user means net or gross if the question is ambiguous
- Always include fiscal year context when citing sales figures
- When citing capacity, state whether it is per machine type or total plant
- Never expose SQL to the user

Tool: `cortex_analyst_text_to_sql` using `SUPREME_DB.ANALYTICS.PRODUCTION_SV`

```
CREATE OR REPLACE AGENT SUPREME_DB.ANALYTICS.SUPREME_AGENT
  AGENT_SPEC = $$
  {
    "instructions": "You are a production and sales analytics assistant for
Supreme Industries.
    Answer questions about net/gross sales by material, machine capacity,
mould usage,
    and pending STO backlog. If 'sales' is ambiguous, ask whether net or
gross.
    Always state the fiscal year for sales figures. State whether capacity
figures
    are per machine type or total plant. Never show SQL.",
    "tools": [{
      "tool_type": "cortex_analyst_text_to_sql",
      "tool_spec": {
        "semantic_model": "SUPREME_DB.ANALYTICS.PRODUCTION_SV"
      }
    }]
  }
  $$;
```

Step 4.2 — Grant Access and Publish to Snowflake Intelligence

CoCo Prompt

Grant `SUPREME_ANALYST_ROLE` the ability to run `SUPREME_AGENT` and query `PRODUCTION_SV` and all tables in `SUPREME_DB.RAW`. Then publish `SUPREME_AGENT` to Snowflake Intelligence with the display

name "Supreme Analytics Assistant" and description: "Ask about sales by material, plant capacity, machine utilisation, and STO backlog."

```
GRANT USAGE ON AGENT SUPREME_DB.ANALYTICS.SUPREME_AGENT TO ROLE
SUPREME_ANALYST_ROLE;
GRANT USAGE ON SEMANTIC VIEW SUPREME_DB.ANALYTICS.PRODUCTION_SV TO ROLE
SUPREME_ANALYST_ROLE;
GRANT SELECT ON ALL TABLES IN SCHEMA SUPREME_DB.RAW TO ROLE
SUPREME_ANALYST_ROLE;

ALTER AGENT SUPREME_DB.ANALYTICS.SUPREME_AGENT SET ACTIVE = TRUE;
```

For users: Snowsight → left sidebar → **Intelligence** → "Supreme Analytics Assistant" → start chatting in plain English.

Step 4.3 — Switch the Underlying LLM Model

The agent can use any Snowflake-hosted LLM. Switch models to tune for speed, reasoning depth, or cost.

Model	Characteristic	Use When
llama3.3-70b	Fast, low-cost	Simple lookups, top-N by material or machine
llama3.1-405b	Largest open model	Complex cross-table, year-over-year comparisons
mistral-large2	Strong structured reasoning	Capacity vs demand gaps, multi-filter analysis
snowflake-arctic	Optimised for SQL accuracy	When SQL precision matters over narrative
claude-3-5-sonnet	Best instruction-following	Nuanced or multi-part business questions

CoCo Prompt

What LLM models are available for Cortex Agents in my Snowflake account region?

CoCo Prompt

Update SUPREME_AGENT to use model llama3.1-405b. Keep all existing instructions and tools exactly as they are.

CoCo Prompt

Run this question through SUPREME_AGENT with model llama3.3-70b and then with llama3.1-405b and show me both answers side by side: "Which materials are at risk of missing their monthly capacity target based on the last 90 days of sales?"

Quick Reference

One-Time Setup

Task	CoCo Prompt
Create DB + schemas + file format	"Create database SUPREME_DB, schemas RAW and ANALYTICS, and a named CSV file format CSV_AUTO in SUPREME_DB.RAW"
Create storage integration	"Create storage integration SUPREME_S3_INT for S3 bucket s3://... using IAM role arn:aws:iam::..."
Get AWS trust policy	"Show me the AWS trust policy JSON for SUPREME_S3_INT"
Create stage	"Create stage S3_STAGE in SUPREME_DB.RAW pointing to s3://... using SUPREME_S3_INT and file format CSV_AUTO"

Load Any File (Every Time)

Task	CoCo Prompt
Load a new file (any schema)	"Load [filename.csv] from @SUPREME_DB.RAW.S3_STAGE — auto-detect schema, create table, load data, show row count"
Load multiple files at once	"Load these files from @S3_STAGE: file1.csv, file2.csv, file3.csv — auto schema, create tables, load all, show summary"
New file — future fiscal year	"Load 26-27 sales.csv from @S3_STAGE into SUPREME_DB.RAW — auto-detect schema, create table if needed, load data"
New STO snapshot	"Load pending_sto_may2026.csv from @S3_STAGE into SUPREME_DB.RAW — auto schema, create table, load"
Schema evolved (new columns)	"A new version of [file.csv] has extra columns. Load it — add new columns to existing table, then load with MATCH_BY_COLUMN_NAME"
Fix load errors	"Show me rows that failed to load from [file.csv] and why"

Semantic View

Task	CoCo Prompt
Describe tables (before building)	"I have these tables in SUPREME_DB.RAW: [describe each]. Confirm your understanding before we build the semantic view."
Create semantic view	"Create PRODUCTION_SV in SUPREME_DB.ANALYTICS on all tables in SUPREME_DB.RAW. Joins: [list]. Metrics: [list]. Business terms: [list]."

Add new table	"I loaded [NEW_TABLE]. It has columns [list] and joins [EXISTING_TABLE] on [KEY]. Add it to PRODUCTION_SV and define metrics: [list]."
Fix wrong answer	"PRODUCTION_SV returned [wrong] for [question]. Correct answer is [X]. Fix the view."
Add VQR	"Add a verified query for '[question]' to PRODUCTION_SV"
Validate	"Test PRODUCTION_SV with 6 representative questions — show SQL and results"

Agent, Chat, and Model

Task	CoCo Prompt
Create agent	"Create Cortex Agent SUPREME_AGENT in SUPREME_DB.ANALYTICS using PRODUCTION_SV. For sales and production managers. Never show SQL."
Grant access + publish to SI	"Grant SUPREME_ANALYST_ROLE access to SUPREME_AGENT. Publish to Snowflake Intelligence as 'Supreme Analytics Assistant'."
List available models	"What LLM models are available for Cortex Agents in my region?"
Switch model	"Update SUPREME_AGENT to use model [model-name]. Keep instructions and tools unchanged."
Compare models	"Run [question] with llama3.3-70b and llama3.1-405b — show both answers side by side"

Starter Questions for Business Users

Share these with your team as opening prompts in Snowflake Intelligence:

Sales

- "What are the top 10 materials by net sales in FY 25-26?"
- "How does FY 25-26 compare to FY 24-25 — which materials grew and which declined?"
- "Which materials have a return rate above 5%?"
- "Show me the monthly sales trend for the top 5 materials over the last year"

Capacity and Plant

- "Which machine type has the highest total daily production capacity?"
- "How many moulds is each machine type configured to run?"
- "Which materials can be produced on more than one machine type?"
- "What is the total plant-wide capacity in units per day?"

Pending STO / Backlog

- "What is the total pending STO backlog today?"
- "Which materials have the highest pending STO quantity?"
- "Which materials have pending STO that exceeds their daily production capacity?"

Cross-Table Analysis

- "Which materials have high sales but constrained capacity — potential supply risk?"
- "For each machine type, what is the net sales volume of materials it produces?"
- "Which materials have pending STO but enough available capacity to fulfil it this week?"
- "Which machine type is most over-utilised based on the last 90 days of sales?"

Every question the agent cannot answer correctly is a permanent improvement opportunity. Use the **"Fix wrong answer"** prompt from the Quick Reference and the correction is instantly available to all users.